

Assessing the Impact of Superspreaders on Epidemic Spread: A Reimplementation of the Fujie-Odagaki Model

Le Van Tuan Anh, Nguyen Bao Phong, Nguyen Minh Quan, Vu Tuan Truong

Hanoi University of Science and Technology (HUST)

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- **SARS Context (2003):** Over 8,000 cases and 700 deaths globally.
- **Superspreaders:** CDC defines as infecting > 10 individuals; causes large cluster surges.
- **Standard SIR Limitations:** Assumes uniform transmission and a well-mixed population.
- **Project Objective:** Reimplement Fujie-Odagaki spatial model to test biological vs. social network hypotheses.

- **Spatial Structure:** Agent-based framework embedded in a continuous 2D space.
- **Methodology:** Replaces deterministic ODEs with stochastic Monte Carlo simulations.
- **Classical SIR Model:**

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \quad \frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

- **Transmission Rule:** Risk is a function of geometric distance r instead of uniform mixing.

- **Environment:** N fixed individuals in an $L \times L$ domain with horizontal periodic boundaries.
- **Normal Individuals:**

$$w_{normal}(r) = \begin{cases} w_0(1 - \frac{r}{r_0})^2, & 0 \leq r \leq r_0 \\ 0, & r > r_0 \end{cases}$$

- **Hypothesis 1 (Strong Infectiousness):**

$$w_{strong}(r) = \begin{cases} w_0, & 0 \leq r \leq r_0 \\ 0, & r > r_0 \end{cases}$$

- **Hypothesis 2 (Hub Model):**

$$w_{hub}(r) = \begin{cases} w_0(1 - \frac{r}{r_n})^2, & 0 \leq r \leq r_n \\ 0, & r > r_n \end{cases} \quad (\text{where } r_n = \sqrt{6}r_0)$$

- **Capacity Normalization:**

$$I_{strong} = \pi w_0 r_0^2 = \frac{\pi w_0 r_n^2}{6} = I_{hub} \implies r_n = \sqrt{6}r_0$$

- **Mechanism:** Discrete-time stochastic steps following $S \rightarrow I \rightarrow R$ with recovery probability $\gamma = 1.0$.
- **Percolation Tracking:** Checks if transmission spans from patient zero (bottom center) to the top threshold.
- **Implementation:** Processed via `main_loop()` in `src/engine.py`, averaged over 1,000 independent runs.
- **Core Framework:** *[Algorithm 1 - Monte Carlo simulation for one parameter setting]*.

Solving Algorithm

Algorithm 1 Monte Carlo simulation for one parameter setting

Require: Superspreader probability λ , population size N , model type M

```
1: Generate  $N$  individuals in the square domain
2: Place patient zero at the bottom center and set its state to infected
3: Mark each individual as a superspreader with probability  $\lambda$ 
4:  $percolated \leftarrow false$ 
5: while at least one individual is infected do
6:    $I_t \leftarrow$  all individuals infected at the current time step
7:    $S_t \leftarrow$  all susceptible individuals
8:   for each infected individual  $i \in I_t$  do
9:     for each susceptible individual  $s \in S_t$  do
10:      Compute the distance  $r$  between  $i$  and  $s$ 
11:      Compute  $w(r)$  according to model type  $M$ 
12:      if a random number in  $[0, 1]$  is less than  $w(r)$  then
13:        Set  $s$  to infected
14:        if  $s$  reaches the top percolation threshold then
15:           $percolated \leftarrow true$ 
16:        end if
17:      end if
18:    end for
19:    Recover  $i$  with probability  $\gamma$ 
20:  end for
21: end while
```

Reference Results and Interpretation

- **Output Metrics:** Tracks percolation probability ($\hat{P}_{perc}$), critical density (ρ_c), cases $\bar{C}(t)$, and speed $v(t)$.
- **Threshold Behavior:** Increasing superspreader fraction λ shifts critical density ρ_c lower (*Figure 1 & Figure 2*).

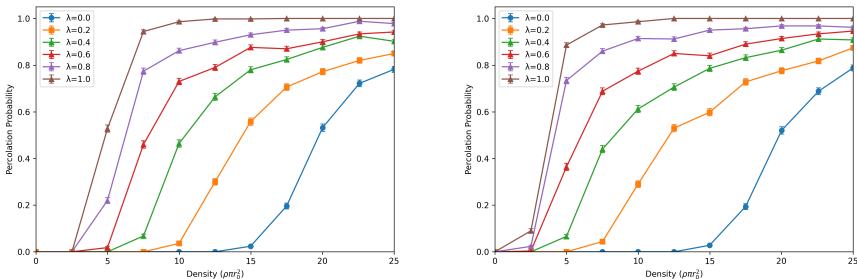


Figure 1: Reimplemented percolation probability curves for the strong infectiousness model (left) and hub model (right).

Reference Results and Interpretation

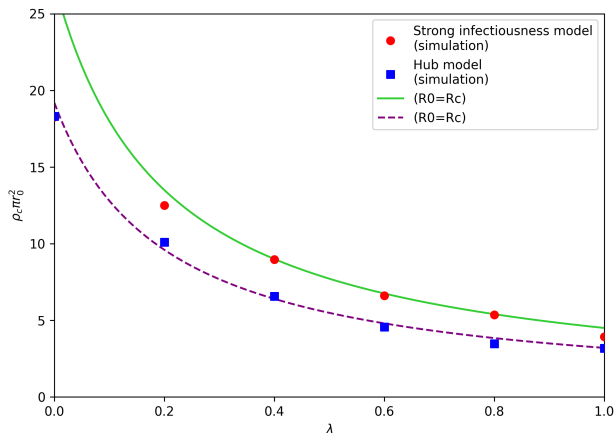


Figure 2: Reimplemented critical density as a function of superspreader fraction.

- **Spatial Routes of Infection:** Visualizing explicit infection paths reveals *why* the hub model spreads more effectively (*Figure 3, Figure 4 & Figure 5*).
 - **Normal** ($\lambda = 0.0$): Short, localized steps only.
 - **Strong** ($\lambda = 0.2$): Dense, localized clusters around superspreaders.
 - **Hub** ($\lambda = 0.2$): Long-range links bridge distant clusters $\Rightarrow$ lower critical density.

Reference Results and Interpretation

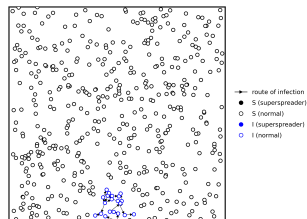


Figure 3: Normal model ($\lambda = 0.0$).

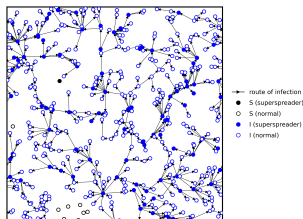


Figure 4: Strong infectiousness ($\lambda = 0.2$).

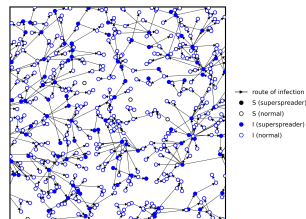


Figure 5: Hub model ($\lambda = 0.2$).

Reference Results and Interpretation

- **Propagation Speed:** Higher superspreader fractions λ drive much faster spatial advancement of the infection front (*Figure 6*).
- **Temporal Concentration:** The outbreak peak is not only larger but also sharper in time, matching the abrupt surges seen in SARS.

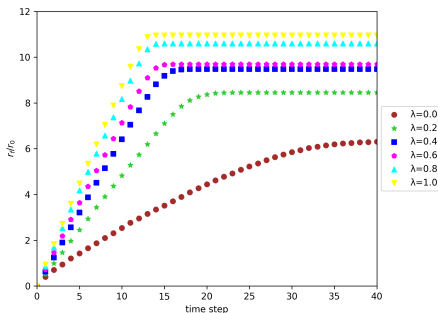


Figure 6: Reimplemented propagation distance over time for varying superspreader fractions.

Reference Results and Interpretation

- **Heterogeneity Insights:** Superspreaders yield long-tailed secondary case distributions and sharper temporal peaks (*Figure 7 & Figure 8*).

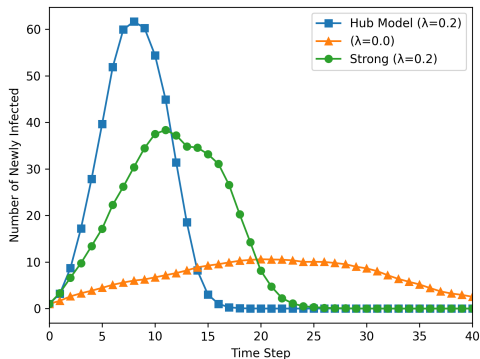


Figure 7: Reimplemented epidemic curves at $\rho\pi r_0^2 = 20$.

Reference Results and Interpretation

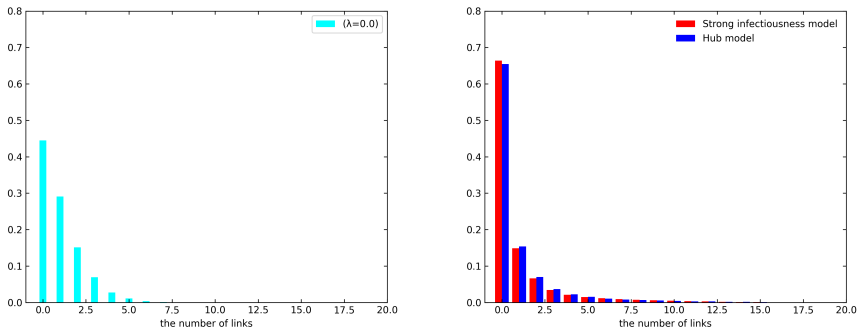


Figure 8: Reimplemented secondary infection distributions at $\rho\pi r_0^2 = 15$.

Reference Results and Interpretation

- **Empirical Validation:** The 2003 Singapore SARS data exhibits the same long right tail as our simulations, with individuals directly infecting 12, 21, 23, and 40 people (*Figure 9*).

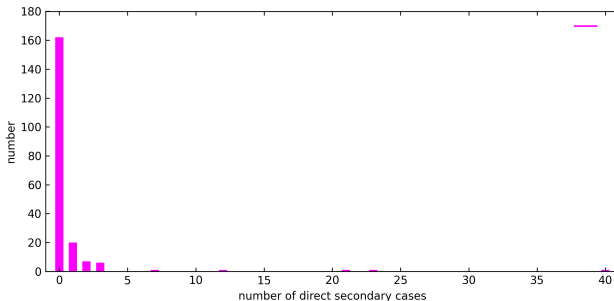


Figure 9: Empirical distribution of secondary cases from the Singapore SARS outbreak.

Reference Results and Interpretation

- **Model Validation:** Hub model demonstrates a superior quantitative match to the empirical 2003 Singapore SARS curve (*Figure 10*).

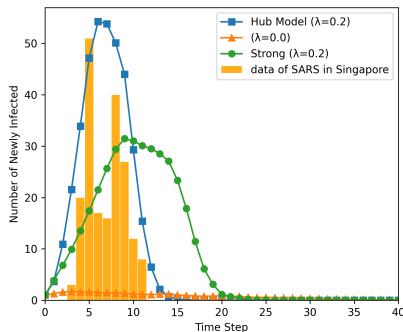


Figure 10: Reimplemented comparison between the simulated epidemic curves and the Singapore SARS epidemic curve.

Robustness of Macroscopic Transition

- **Structural Robustness:** Agent-based percolation thresholds strictly adhere to the continuous theoretical curve $R_0 = R_c$ across all $\lambda \in [0, 1]$ (*Figure 2*).
- **Theoretical Foundation:** Outbreak boundaries are governed by the critical reproductive equation:

$$R_0(\lambda) = \frac{\rho}{w_0} \left[\lambda \int w_{ss}(r) 2\pi r dr + (1 - \lambda) \int w_n(r) 2\pi r dr \right] = R_c$$

Robustness of Macroscopic Transition

- **Model Baseline Thresholds:** Fixed constants derived from continuous spatial epidemic theory:

$$R_c = 4.5 \text{ (Strong Model)} \quad \text{and} \quad R_c = 3.2 \text{ (Hub Model)}$$

- **Geometric Normalization:** The scaling factor $r_n = \sqrt{6}r_0$ ensures identical total outbreak capacity, making the superiority of the hub model a robust geometric finding rather than parameter tuning.

- **Local Finite-Difference Sensitivity:** Formulated to measure system response to input parameter variations (θ):

$$S_{\theta,Y}(\theta_j) \approx \frac{Y(\theta_{j+1}) - Y(\theta_j)}{\theta_{j+1} - \theta_j}$$

- **High Sensitivity to Initial Superspreaders:** Introducing an initial fraction of superspreaders ($\lambda : 0.0 \rightarrow 0.2$) causes a steep drop in critical density:

$$S_{\lambda,\rho_c} \approx -41.10 \text{ (Hub Model)} \quad \text{and} \quad S_{\lambda,\rho_c} \approx -28.87 \text{ (Strong Model)}$$

Sensitivity to Heterogeneity & Noise

- **Statistical Stability against Stochastic Noise:** Negligible standard error bars prove the sharp phase transition is immune to simulation noise (*Figure 1*).
- **Monte Carlo Standard Error:** Calculated over $K = 1000$ independent trials with a maximum bounds of ≈ 0.0158 :

$$SE(\hat{P}_{perc}) = \sqrt{\frac{\hat{P}_{perc}(1 - \hat{P}_{perc})}{K}}$$

THANK YOU!